

KUNAL M K

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ABOUT ME

Final-year Computer Science undergraduate focused on AI-integrated systems, machine learning pipelines, and backend engineering. Experience spans geospatial ML, LLM-based multi-agent systems, blockchain security, and systems-level desktop programming in C#/.NET. Proficient in Python, Java, and C++, with strong foundations in data structures, algorithms, and system design.

EDUCATION

PES College of Engineering, Mandya

2023 – Present

B.E. in Computer Science and Engineering | Final Year | CGPA: 8.68

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Object-Oriented Programming

Sankalpa PU College

2021 – 2022

Pre-University Education (PCMB)

Carmel Academy ICSE

2020

Secondary School Certificate — 10th Grade

SKILLS

Languages: Python, Java, C++, C, SQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Object-Oriented Design, System Design, DBMS, Operating Systems, Computer Networks

AI / Machine Learning: LLMs, LangChain, RAG, Multi-Agent Systems, Prompt Engineering, Hugging Face, Scikit-learn, LightGBM, Random Forest, SVM, NumPy, Pandas, Matplotlib, FAISS, ChromaDB, Groq API

Backend & Development: REST APIs, FastAPI, Streamlit, Git, GitHub, GitHub Actions (CI/CD), Jupyter Notebook, VS Code

Blockchain: Stellar, Soroban, Smart Contracts, IPFS, SHA-256, On-chain Data Anchoring

Cloud: Google Cloud, Firebase

PROJECTS

Flood Risk Zonation System

2025

Python • Scikit-learn • GeoPandas • Rasterio • Streamlit • Folium • NASA SRTM

- Built a scalable ML geospatial application processing live satellite data (NASA SRTM 30m DEM) across distributed grid-cell computations at 500m resolution, generating real-time flood risk maps for any global region.
- Designed a 10-feature extraction pipeline using Random Forest with 5-fold cross-validation (AUC ≈ 0.97); engineered the full data ingestion → training → inference → export stack end-to-end.
- Deployed on Streamlit Cloud with interactive visualizations, automated PDF report generation, and multi-layer map overlays — delivering a production-grade user experience.

MetroMeet AI — Intelligent Metro-Based Group Meetup Planner

2026

HTML5 • CSS3 • JavaScript • Node.js • Express.js • Anthropic Claude API • HTML5 Canvas

- Modeling the Bangalore Namma Metro network as a weighted graph and applying Dijkstra's algorithm with automatic route reconstruction for optimal multi-user routing.
- Designing a multi-objective optimization engine balancing travel time, fairness across users, interchange penalties, and station attractiveness scoring to recommend the best meetup location.
- Building a custom HTML5 Canvas map renderer with real-time zoom/pan and route highlighting, an autocomplete search index, and an AI assistant for contextual route explanations.

Com-paw-nion — AI-Powered Interactive Desktop Companion

2026

C# • .NET 8 • WPF • SkiaSharp

- Designing and building a modular Windows desktop companion platform with a custom SkiaSharp-based rendering pipeline and transparent overlay windows, structured using Clean Architecture across 6 core projects.
- Engineering a custom animation engine (animation graphs, state transitions, event-driven playback) alongside an allocation-free procedural animation system for idle motion such as breathing and body movement.
- Architecting a physics-based interaction layer and plugin system to support extensible companions, behaviors, and AI personalities on top of a dependency-injected, testable codebase.

ShieldChain — AI Smart Contract Security & On-Chain Audit

2024

Python • AI Agents • Stellar/Soroban • IPFS • SHA-256

- Designed a multi-agent AI pipeline performing static, semantic, and dynamic analysis on smart contracts, reducing manual audit time by 60%+ through automated vulnerability detection across 5 risk dimensions.
- Implemented end-to-end system architecture — AI analysis agents → hashing layer → decentralized storage — demonstrating the ability to design and ship complex, multi-component systems.

EXPERIENCE

Lead — Google Developer Group On Campus (GDGoC), PESCE Mandya

Aug 2025 – Jul 2026

- Led a team of 15+ members to plan and deliver 8+ technical events and workshops focused on AI and Cloud, increasing community participation by 40% through targeted outreach and hands-on learning sessions.

Student Placement Coordinator — Dept. of Training & Placement, PESCE Mandya

2023 – Present

- Coordinated 8+ placement drives with 15+ recruiting companies, managing end-to-end interview logistics for 200+ students and reducing placement cycle time by streamlining recruiter-student communication.

Club Head, Matrixz CSE | Core Member, Institution Innovation Council (IIC)

2023 – 2026

- Founded and grew Matrixz CSE to 100+ members, running regular coding contests and DSA workshops; organized 5+ startup and innovation events through IIC, engaging 150+ students across departments.

CERTIFICATIONS & ACHIEVEMENTS

- Qualified HackWithInfy 2026 Rounds 1 and 2 by solving advanced DSA problems, earning a Pre-Placement Interview slot at Infosys for SP/DSE roles.
- Google Cloud Skill Boost badges: Generative AI Fundamentals, Google Cloud Fundamentals.
- Regularly solve algorithmic problems on LeetCode and Codeforces spanning arrays, graphs, dynamic programming, and greedy algorithms.